ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/346955420

A satellite collision avoidance system based on General Regression Neural Network

Conference Paper · December 2020

DOI: 10.1109/BDCAT50828.2020.00004

CITATIONS
3

READS
442

5 authors, including:

Md.Riftabin Kabir
North South University
3 PUBLICATIONS **33** CITATIONS

SEE PROFILE

Tarek Ibne Faysal
North South University
3 PUBLICATIONS **4** CITATIONS

SEE PROFILE

Md. Sazzad Hossain
North South University
8 PUBLICATIONS **12** CITATIONS

SEE PROFILE

Jannatun Naima Shorno
North South University
1 PUBLICATION **3** CITATIONS

SEE PROFILE

All content following this page was uploaded by Shahnewaz Siddique on 25 June 2022.

The user has requested enhancement of the downloaded file.

A satellite collision avoidance system based on General Regression Neural Network

1st Md. Riftabin Kabir[1]
North South University
Dhaka, Bangladesh

2nd Tarek Ibne Faysal[1]
North South University
Dhaka, Bangladesh

3rd Md. Sazzad Hossain[1]
North South University
Dhaka, Bangladesh

4th Jannatun Naima Shorno[1]
North South University
Dhaka, Bangladesh

5th Shahnewaz Siddique [2]
North South University
Dhaka, Bangladesh

***Abstract*—Continuous launching of new satellites and increasing numbers of space missions is making space a congested environment. Collision with space debris or other satellites is now a real problem for satellites with the problem more acute in highly trafficked orbits. Thus, mission operators and space agencies are in need of high accuracy collision avoidance systems for spacecrafts and satellites. This paper focuses on tackling the satellite collision problem by implementing a collision avoidance system using neural networks and relevant machine learning techniques. The primary model is based on General Regression Neural Network (GRNN) and the secondary models are based on Artificial Neural Networks (ANN), Random Forest Regression & Support Vector Regression techniques. The dataset used in this paper is collected from the ESA (European Space Agency) which contains the events of risk assessment or in other words, Conjunction Data Messages (CDM). The proposed collision avoidance system predicts the collision risk percentage between target (a satellite of interest) and chaser (space debris or another satellite) objects. The predicted risk enables the target to maneuver accordingly and ultimately avoid collision with the chaser object. The GRNN algorithm uses lazy learning which does not require iterative training and makes predictions based on stored parameters. The training data has been normalized before applying the algorithm as GRNN network is sensitive to high deviation among input features. However, the GRNN model predicts the risk of collision between the target & the chaser object with an MSE (Mean Squared Error) of 11% which means the model predicts the risk of collision with 89% accuracy and this 89% risk can give enough confidence factor to the concerned authority to take necessary evasive maneuvers. This is reliable enough and lower than other models' MSE to consolidate the fact that the GRNN model is best fit for our dataset.**

***Index Terms*—Space Systems, Satellites, Spacecrafts, Neural Network, GRNN, Machine Learning, Collision Avoidance**

I. Introduction

Despite the current physics-based models for predicting orbits and implementing collision avoidance systems in space, a significant number of collisions still occur, leaving space with increasing amount of space debris [9]. These space debris are a serious hazard causing, and will cause in the future, collisions and disrupting the safety of space missions and satellites. 2007 witnessed the destruction of Fengyun-1C, in 2009 collision occurred between the Iridium-33 and the Cosmos-2251 spacecrafts, and in 2012 the explosions of the Briz-M [7] significantly increased the amount of debris floating in space. These space debris represent great potential danger to all spacecrafts, satellites and other space entities. According to the European Space Agency, approximately 128 million objects are floating around in space which measures between 1 mm to 1 cm. The number is 900,000, when it comes to objects measuring between 1 cm to 10 cm and last but not the least, the objects measuring greater than 10 cm are 34,000 in number and rising [4]. Starting from LEOs (Low Earth Orbits) to HEOs (High Earth Orbits), these debris are distributed in varying amounts [13] which can prove to be the major causes of collisions and accidents for the satellites and spacecrafts operating in these regions. Moreover, LEO is considered to be the 'Orbital Graveyard'. The amount of large debris has increased up to 70% in LEO just due to the Fegyun-1C destruction in 2007 and the collision of the American and Russian spacecrafts in 2009. The significant increase of space debris is a great threat to the safe movement and operations of the satellites and other spacecrafts in Low Earth Orbits and also in High earth orbits [8].

This research focuses on mitigating the risks of collisions and accidents discussed above. Unlike other conventional collision avoidance models, the model which is proposed in this paper is developed with General Regression Neural Networks (GRNN) which predicts the risk measure of collision of the target and the incoming space entities. With the estimated risk level, the spacecraft or satellite can maneuver accordingly to avoid the collision with the incoming objects (space debris). The proposed model has two aspects where one covers the risk factors considering only the high risk data and the other gives a prediction based on the overall data of risk assessment. To the best of our knowledge, there are no existing works of neural collision avoidance system for spacecrafts and satellites which are done by machine learning algorithms like GRNN and with the dataset of real world Conjunction Data Messages (CDM) prepared by the European Space Agency [3]. The study does not calculate the new orbit after evasion of the debris

[1]Research Assistant, Department of Electrical & Computer Engineering.
[2]Assistant Professor, Department of Electrical & Computer Engineering, IEEE member.

978-0-7381-2396-7/20/$31.00 ©2020 IEEE
DOI 10.1109/BDCAT50828.2020.00004

because the purpose of this study is to predict the probability of collision only. Maneuvering after evasion is a great scope for future studies and improvement of this paper. However, The model can be considered as one of the advanced approach to achieve a neural collision avoidance system for satellites and spacecrafts with adaptive maneuvering measures based on the risk analysis.

II. Related Works

Finding previous work which focuses on machine learning approach regarding satellite collision avoidance system is exceptionally hard currently since it is a very new field of work. Looking into the traditional way of collision avoidance of satellites developed by European Space Agency Office [7] is a good start. Although machine learning approach regarding aircraft collision avoidance is available, this paper does not include those studies as spacecrafts require totally different maneuvering compared to standard airplanes.

Collision avoidance systems have become important components in air and space traffic systems even more so after the fatal destruction of Fengyun-1C in 2007, the Iridium-33/ Cosmos-2251 collision in 2009, and the Briz-M explosions of 2012 [7], which eventually resulted in more debris in the space atmosphere. European Space Agency's Space Debris Office has some collision avoidance services which is being currently used to compensate the issue. Their activities includes model development, re-entry risk assessments and create development standards. Conjunction Data Messages (CDMs) provided by Space Control Squadron [7] are retrieved and analysed to estimate collision probability. The estimates are further assessed and maneuvers optimized considering various constraints. As the key data are known with limited precision, the trajectories of both objects have uncertainties around the time of closest approach. Therefore, prime criterion of any collision avoidance strategy is when to execute an avoidance maneuver and whether to execute it or not. ESA's ARES [10], [2] tool is used to determine reasonable reaction thresholds. It allows the system to figure out collision risks above levels selected by the user in regards of spacecraft size, age, quality [7].

Airborne Collision Avoidance System X (ACAS X) family formulates the aircraft collision avoidance problem as a Markov Decision Process (MDP) and uses Dynamic Programming to optimize the system [5]. Due to the high dimensionality of the state space, it results in a large table. Even though down-sampling the table after dynamic programming reduces the size of the score table, the required space is much more than the legacy hardware processing capabilities. Recent algorithms exploits the natural symmetry of the score table to remove redundancy within the table. Decision trees also offer a way of organizing the data into trees to remove redundancy and increase compression. Linear regression for small dataset and Support Vector Machine (SVM) are also popular for compressing the data which is to be stored. Neural networks can serve as a robust function approximator and if trained efficiently can also represent large dataset. As only the network parameters need to be stored, it can reduce the required storage space by a factor of 1000 [5].

III. Methodologies

The process flow of data pre-processing, model creation, model training and prediction is demonstrated in Fig. 1.

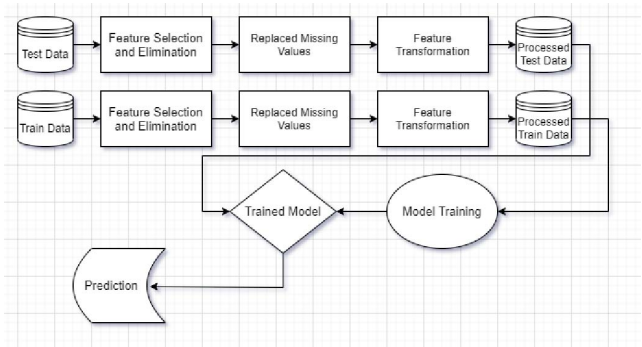

Fig. 1. Methodologies Diagram

A. *Data*

The primary dataset is collected from the Kelvins - ESA Advanced Concepts Competition website [3]. This particular dataset is as complex as it can get with varied degree of sophistication in terms of both analytical and statistical significance. The dataset consists of real world Conjunction Data Messages (CDM) which are prepared at the ESA. The CDMs are created after thoroughly analyzing and filtering hundreds of alerts which are issued every week corresponding to possible close encounters between a satellite and another space object. The data can be better visualized with the help of Fig. 2 [7].

With about 12 rows/CDMs per close encounter, there are 162634 instances and 13154 unique events in the training set and there are 24484 instances and 2167 unique events with about 11 rows/CDMs per close encounter, in the testing set. A maneuver to avoid a collision is planned at least 2 days prior to the approach and that is why no CDMs exist withing 2 days of the time to closest approach from which we can learn. Primarily each CDM in the dataset had 103 characteristics/ attributes. For each collision, a time series of CDMs are considered as a single collision event. Among the 103 attributes the uniquely named ones for a clear understanding of the major characteristics of the dataset in concern are demonstrated in Table I.

However there are some shared attribute definitions between the chaser & the target object. Such as-

1. x_sigma_rdot
2. x_sigma_n
3. x_cn_r
4. x_cn_t
5. x_cndot_n
6. x_sigma_ndot
7. x_cndot_r
8. x_cndot_rdot

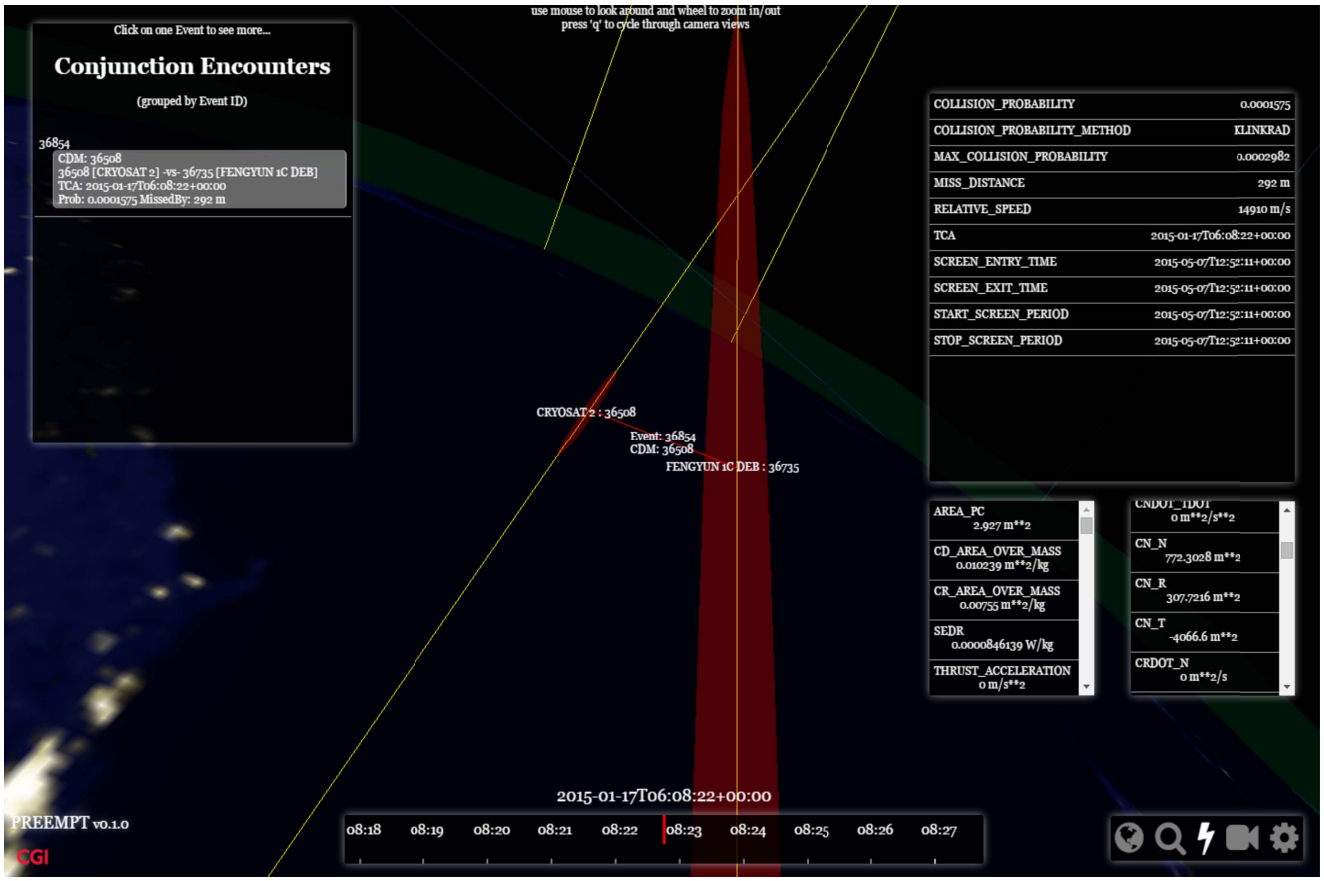

Fig. 2. Conjunction visualization with details and object positions and covariance ellipsoids(Cryosat-2 encounter with a Fengyun-1C debris object). Courtesy of [7].

9. x_position_covariance_det
10. x_cd_area_over_mass
11. x_rcs_estimate
12. x_recommended_od_span
13. x_residuals_accepted
14. x_time_lastob_end
15. x_time_lastob_start
16. x_weighted_rms etc.

The complete set of attributes and their definitions can be found at the Kelvins ESA Advanced Concept Website for deeper understanding [3]. All the attributes are numerical in nature except the c_object_type.

Among the 103 attributes, 'AP', 'c_object_type', 'SSN' and the shared ones between the target and the chaser, were discarded. The final dataset for the study consists of 18 attributes including the target class with instances accumulating upto 1,87,118 (one lacs eighty seven thousand & one hundred and eighteenth) combining the training and the testing data.

B. *Pre-processing*

Due to the sheer complexity of the dataset it was strenuous to work with it. For the best outcome of the study and the data itself, the total dataset had to go under scrutinizing manipulation and pre-processing through various methods. The basic stages before proceeding to train the model are discussed below:

*1) **Feature Selection & Elimination**:* After an empirical study of the attributes of the CDMs and risk factors, the 18 attributes were selected after excluding the above mentioned attributes.

*2) **Replacing Missing and Null Values**:* After the feature selection, the whole dataset was thoroughly checked for missing and null type values and were replaced by the standard mean value of the column for a specific attribute.

*3) **Attribute Transformation**:* As mentioned before, the data has been normalized before applying the algorithm as GRNN is sensitive to high deviation among input features. After cleaning the data, the values of the attributes were greatly varied because of the complex structure and characteristics of the dataset. For mitigating this large scale variation, the dataset standardization and transformation of the attributes were done. First the min & max values of the attributes were found and then the dataset was scaled using MinMaxScaler of scikit-learn. Then the whole dataset was transformed using fit_transform method of scikit-learn library. The transformed

TABLE I
ATTRIBUTE DEFINITIONS AND TYPES

Attribute Name	Definition
risk (target attribute)	Self-computed value at the epoch of each CDM [base 10 log]
event_id	Time interval between CDM creation and time-of-closest approach [days]
mission_id	Identifier of mission that will be affected
max_risk_estimate	Maximum collision probability obtained by scaling combined covariance
max_risk_scaling	Scaling factor used to compute maximum collision probability
miss_distance	Relative position between chaser & target at tca [m]
relative_speed	Relative speed between chaser & target at tca [m/s]
relative_position_n	Relative position between chaser & target: normal (cross-track) [m]
relative_position_r	Relative position between chaser & target: radial [m]
relative_position_t	Relative position between chaser & target: transverse (along-track) [m]
relative_velocity_n	Relative velocity between chaser & target: normal (cross-track) [m/s]
relative_velocity_r	Relative velocity between chaser & target: radial [m/s]
relative_velocity_t	Relative velocity between chaser & target: transverse (along-track) [m/s]
c_object_type	Object type which is at collision risk with satellite
geocentric_latitude	Latitude of conjunction point [deg]
azimuth	Relative velocity vector: azimuth angle [deg]
elevation	Relative velocity vector: elevation angle [deg]
F10	10.7 cm radio flux index
AP	Daily planetary geomagnetic amplitude index
F3M	81-day running mean of F10.7 (over 3 solar rotations)
SSN	Wolf sunspot number

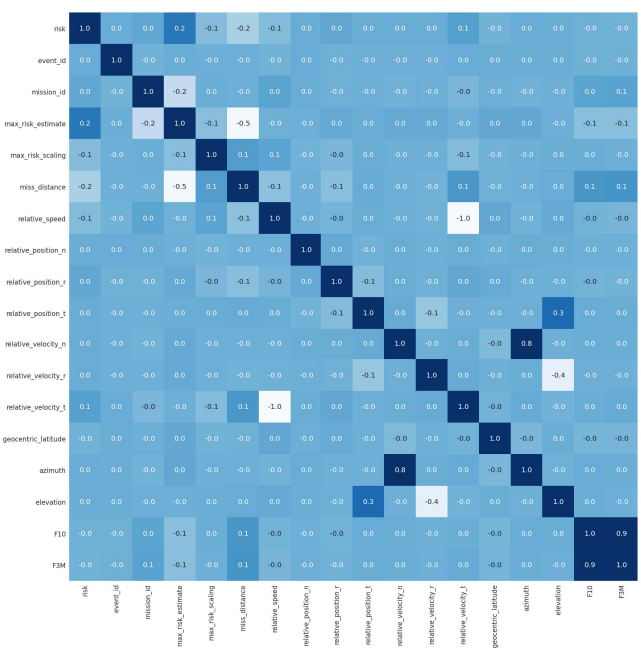

Fig. 3. Correlation Matrix

dataset can be some what visualized by Fig. 4.

After the above three stages of cleaning and processing, the dataset was ready for the training of the models.

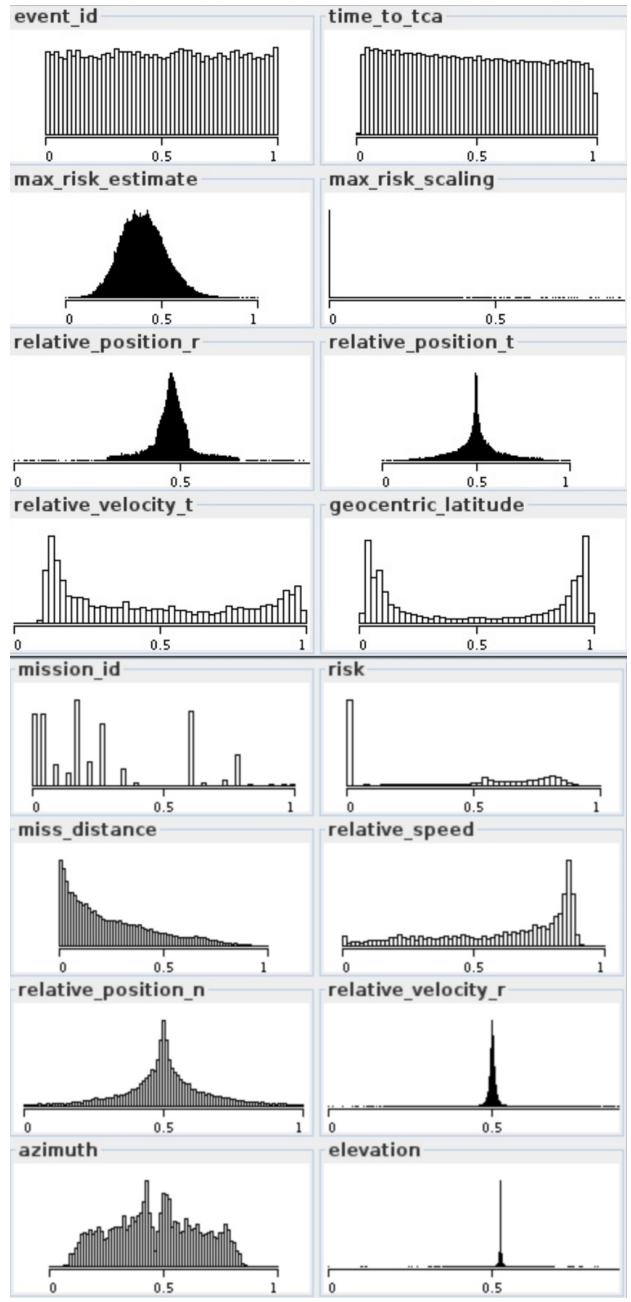

Fig. 4. Transformed Data

C. Algorithms & Methods

1) General Regression Neural Network: The primary and main model of our research is based on GRNN. It is a variation to radial basis neural networks. The network architecture is demonstrated through Fig. 5 [14]. It makes predictions based on stored parameters, so it doesn't need any iterative training. GRNN is also sensitive to highly deviated input features. The most significant parameter of this algorithm is 'Standard

TABLE II
MODEL SUMMARY OF ANN

Layer	Output Shape	Param#
dense_4 (Dense)	(None, 2048)	204800
dense_5 (Dense)	(None, 4096)	8392704
dense_6 (Dense)	(None, 512)	2097664
dropout_1 (Dropout)	(None, 512)	0
dense_7 (Dense)	(None, 1)	513

TABLE III
PARAMETER CONFIGURATION

Total Params	10,695,68
Trainable Params	513
Non-trainable Params	10,695,168

Deviation' for Probability Density Function. While training the model, **100** was used as the **std parameter** value for the algorithm in concern. The algorithm is mathematically represented in equation 1 where Y(x) is the prediction value of input x, y_k is the activation weight for the pattern layer neuron at k and K(x,x_k) is the Radial basis function kernel (Gaussian kernel).

$$Y(x) = \frac{\sum_{k=1}^{N} y_k K(x, x_k)}{\sum_{k=1}^{N} K(x, x_k)} \tag{1}$$

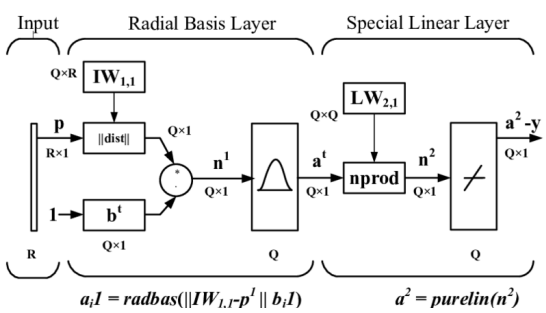

Fig. 5. GRNN Architecture. Courtesy of [14].

*2) **Artificial Neural Networks**:* The second algorithm that was used for the research was ANN. The model was compiled using **adam** as optimizer value and **mean_squared_error** as loss value. The activation units in the layer was **relu**. The model summary is demonstrated by Table II and Table III. To train the model, random 20% of the training data was used for the validation set. Finally, the model was trained using NVIDIA RTX 2080TI GPU.

*3) **Random Forest Regressor**:* One of the techniques of machine learning to make more accurate predictions than any individual model is to combine the predictions from multiple machine learning algorithms. Random Forest Regressor operates by constructing a multitude of decision trees and give the output of the class that is the mean prediction (regression) of the individual trees. The network architecture is demonstrated by Fig. 6 [12]. It uses **Bootstrap Aggregation** in which Bootstrap refers to random sampling with replacement. For each Decision tree, the algorithm calculates a nodes importance using Gini importance. For training, the value of number of estimator was 100. The Mathematical representation of the MSE formula is demonstrated via equation 2.

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(f_i - y_i)^2 \tag{2}$$

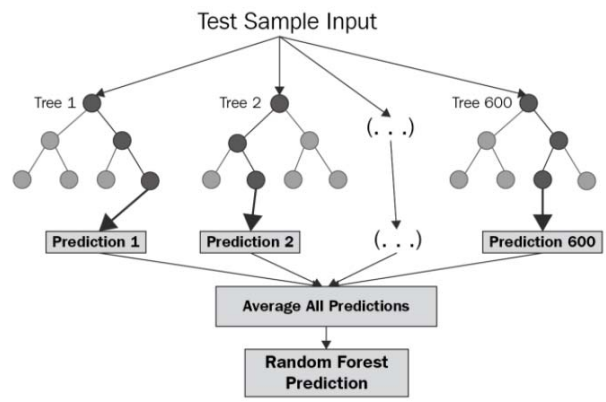

Fig. 6. Random Forest Regression. Courtesy of [12].

*4) **Support Vector Regression**:* SVR is an extremely useful algorithm because of its error tolerance tuning feature. It can be done via both acceptable error margin and error rate margin tolerance tuning. The SVR model can visualized by Fig. 7 [11].

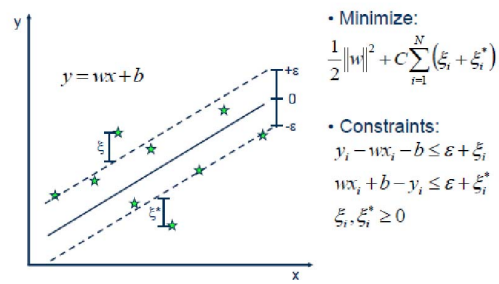

Fig. 7. Support Vector Regression. Courtesy of [11].

Here, epsilon is the margin of tolerance, C is a constant, w is weight and b refers to bias. The two parameters that are crucial to the model are **C** and **epsilon**. For C, the value was 1.0 and the value of epsilon was 0.2 . These two parameters control the error margin tolerance of our model and thus gives a benefit of control over the model's error value.

*5) **Risk Calculation**:* To give an understanding of risk calculation with the attributes of the dataset, the current collision avoidance system technique can be discussed. For example, the ISS takes decision of maneuvering based on the

probability of collision. The probability of collision can be explained as the probability that the debris object will intercept the sphere of radius R during the encounter [1], [6]. In Fig. 8 [1], a set of trajectories for the debris and ISS given by $\vec{r_{do}}$ and $\vec{r_{so}}$ respectively, where $\vec{e_d}$ and $\vec{e_s}$ are the uncertainty vectors for the debris and ISS.

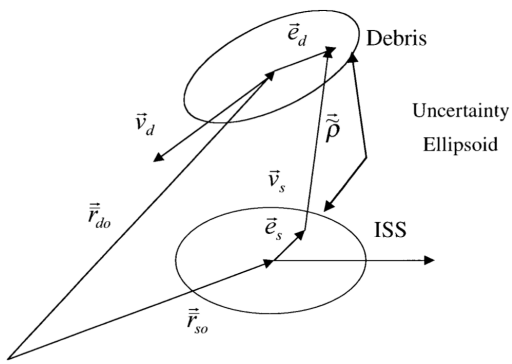

Fig. 8. Geometric view of space encounters. Courtesy of [1].

At an instant time, the probability of collision is calculated via equation 3.

$$P_c = \frac{1}{(2\prod)^{\frac{3}{2}}(detP)^{\frac{1}{2}}} \int_v exp(-A)dV \tag{3}$$

Here the integral is over the sphere of R radius. The P_c stands for probability of collision, P is co-variance and the A is matrix of partial derivatives of the measurements with respect to the initial state. With the slightest change in co-variance P, there may be fast changes in P_c. If PDF is considered constant over the collision sphere, a better approximation of P_c can be achieved [1] [6].

This study tries to introduce a new technique to predict the risk of collision which is done by many state of art technologies and methods like the above discussed system. However, it does not necessarily includes or draws comparisons to the current state of the art technologies. This study focuses on establishing a whole new domain of technology and method to predict the risk of collisions which will help to take evasive maneuvers accordingly.

In this study, the target is to predict the risk with minimum error and highest accurate approximation. Regression models of Neural Networks are mainly used to obtain this target as much as possible with a goal for further optimization in error minimization in predicting the risk of collision.

IV. Result Analysis & Evaluation

The performance measure of the trained models is MSE (Mean Squared Error). The MSE (Mean Squared Error) and MAE (Mean Absolute Error) values of the four trained models are shown at Table IV.

MSE is inversely proportional to model's performance. The lower the MSE, the better the model's performance. In Table IV it can be observed that General Regression Neural Networks model has the lowest MSE among all the 4 models in this study which is 0.11 and the prediction graph is visualized at Fig. 9.

TABLE IV
MSE of Model

Model	MSE Value	MAE Value
General Regression Neural Network	0.11	0.32
Artificial Neural Networks	0.13	0.33
Random Forest Regressor	0.19	0.34
Support Vector Regression	0.14	0.34

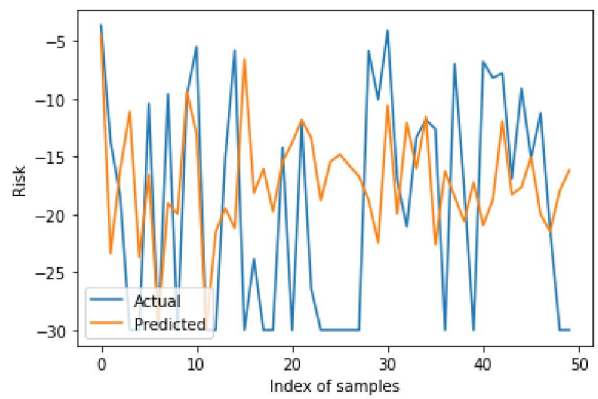

Fig. 9. Prediction Graph - GRNN

After GRNN, the second lowest MSE is of the model ANN. It has 0.13 as MSE and 0.33 as MAE value. Support Vector Regression has the third lowest MSE which is 0.14 and the prediction graph of SVR can be seen at Fig. 10. Lastly, with the highest MSE value of 0.19 the Random Forest Regressor could not perform well compared to the other three.

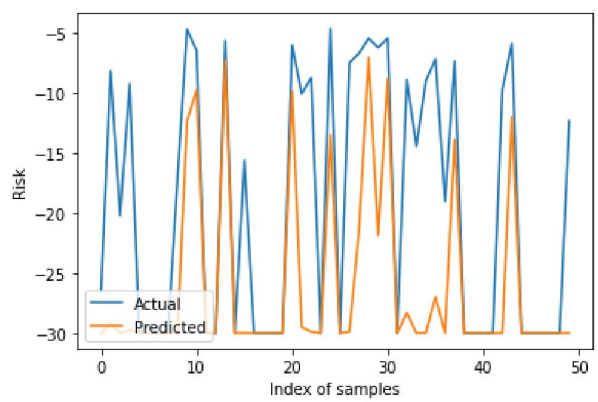

Fig. 10. Prediction Graph - SVR

The prediction graphs of the GRNN and the SVR model are generated by taking randomly selected 50 samples from the test data (actual) and thus the actual graph differs in the two

graphs. The negative y-axis represents the self-computed risk value at the epoch of each CDM [base 10 log]. The closer the value to positive the lesser risk it represents and vice-versa.

The MSE and the MAE values indicates the models performances and depicts the sole purpose of the study, risk-prediction. The MSE value of 0.11 of the GRNN model means that the model predicts the risk of collision with 89% accuracy. To be more precise, there will be 89% risk of collision between the chaser and the target. This is high and alarming enough to take necessary measures to mitigate the risk and eventually avoid the collision. The all other models also denotes a 80% + on average risk factor of collision which can be really helpful to make decisions regarding the evasive maneuvers.

As it is a matter of space collisions and sophisticated calculations, even the slightest measurement error can lead to disastrous situations. That is why the error margin has to be considered carefully, no matter how small the difference is. With all aspects in focus, the GRNN is the best performing model among the four models. However, it is impeccable that, the GRNN model is the best suited one for the sole purpose of the study, which is predicting the risk of collision with minimum error and ensure safer space missions.

V. Conclusion

With increasing space missions and commercial space flights, the amount of debris in LEO and HEO will only keep increasing. As a result, it will be increasingly difficult to operate safe space missions and avoid satellite collision mishaps. The collision avoidance task can become gradually convenient by employing modern deep learning techniques and this particular study is an example of that. The risk prediction model can be used to estimate the incoming risk factors and swiftly take necessary precautions to avoid any hazards. It can be extended to analyze the risk events furthermore and guide the satellites or spacecrafts to maneuver accordingly. The proposed model and the whole study itself can be a path to many upcoming precise systems that will make space collision free and pave the way to safer space missions in general.

References

[1] Alfriend, K.T., Akella, M.R., Frisbee, J., Foster, J.L., Lee, D.J., Wilkins, M.: Probability of collision error analysis. Space Debris **1**(1), 21–35 (1999)

[2] Dominguez-Gonzalez, R., Sanchez-Ortiz, N., Krag, H., Gelhaus, J.: Analysis of uncertainties of catalogued orbital data for the update of the ESA DRAMA ARES tool. In: Proceedings of the 63rd International Astronautical Congress (2012)

[3] ESA: Kelvins European Space Agency Advanced Concepts Competition . available: https://kelvins.esa.int/

[4] ESA: Space debris by the numbers available: https://www.esa.int/

[5] Julian, K.D., Kochenderfer, M.J., Owen, M.P.: Deep neural network compression for aircraft collision avoidance systems. Journal of Guidance, Control, and Dynamics **42**(3), 598–608 (2019)

[6] Klinkrad, H., Alarcon, J., Sanchez, N.: Collision avoidance for operational ESA satellites. In: 4th European Conference on Space Debris. vol. 587, p. 509 (2005)

[7] Merz, K., Virgili, B.B., Braun, V., Flohrer, T., Funke, Q., Krag, H., Lemmens, S., Siminski, J.: Current collision avoidance service by ESA's space debris office. In: Proceedings 7th European Conference on Space Debris, Darmstadt, Germany. pp. 18–21 (2017)

[8] NASA: Space debris available: https://www.nasa.gov/

[9] Peng, H., Bai, X.: Artificial neural network–based machine learning approach to improve orbit prediction accuracy. Journal of Spacecraft and Rockets **55**(5), 1248–1260 (2018)

[10] Sánchez-Ortiz, N., Domínguez-González, R., Krag, H., Flohrer, T.: Impact on mission design due to collision avoidance operations based on tle or csm information. Acta Astronautica **116**, 368–381 (2015)

[11] Sayad, S.: Support vector machine available: www.saedsayad.com/support_vector_machine_reg.htm

[12] TDS: "Random forest and its implementation". available: https://towardsdatascience.com/random-forest-and-its-implementation-71824ced454f

[13] Wang, Y., Gurfil, P.: Dynamical modeling and lifetime analysis of geostationary transfer orbits. Acta Astronautica **128**, 262–276 (2016)

[14] Yang, S., Ting, T., Man, K.L., Guan, S.U.: Investigation of neural networks for function approximation. Procedia Computer Science **17**, 586–594 (2013)

View publication stats